

CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · PYQ

Chapter 9: Matrices

Previous year board questions, topic-wise: 8 questions

The tag after each question gives the board examination year and the marks. Solutions are in a separate PDF with the same question numbers.

1 Multiplication of matrices

1. If $A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$, find A^2 . (a) $\begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$ (b) $\begin{bmatrix} 0 & 0 \\ 1 & 1 \end{bmatrix}$ (c) $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ (d) $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ ICSE 2025, 1 mark
2. Given $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$, $B = \begin{bmatrix} 2 & 1 \\ 4 & 2 \end{bmatrix}$ and $C = \begin{bmatrix} -5 & 1 \\ 7 & -4 \end{bmatrix}$, find (a) $A + C$ (b) $B(A + C)$ (c) $5B$ (d) $B(A + C) - 5B$. ICSE 2025, 4 marks
3. If $A = \begin{bmatrix} 2 & 2 \\ 0 & 2 \end{bmatrix}$ and $A^2 = \begin{bmatrix} 4 & x \\ 0 & 4 \end{bmatrix}$, find x . (a) 2 (b) 4 (c) 8 (d) 10 ICSE 2024, 1 mark
4. Given $A = \begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$, $C = \begin{bmatrix} 4 & 1 \\ 1 & 5 \end{bmatrix}$ and $I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$, find $A(B + C) - 14I$. ICSE 2023, 3 marks
5. If $A = \begin{bmatrix} 3 & 0 \\ 5 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} -4 & 2 \\ 1 & 0 \end{bmatrix}$, find $A^2 - 2AB + B^2$. ICSE 2020, 3 marks

2 Solving matrix equations

6. $A = \begin{bmatrix} x & 0 \\ 1 & 1 \end{bmatrix}$, $B = \begin{bmatrix} 4 & 0 \\ y & 1 \end{bmatrix}$ and $C = \begin{bmatrix} 4 & 0 \\ x & 1 \end{bmatrix}$. Find x and y if $AB = C$. ICSE 2024, 4 marks
7. If $\begin{bmatrix} 2 & 0 \\ 0 & 4 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 2 \\ -8 \end{bmatrix}$, find x and y . (a) 1, -2 (b) -2, 1 (c) 1, 2 (d) -2, -1 ICSE 2023, 1 mark
8. $A = \begin{bmatrix} x & 3 \\ y & 3 \end{bmatrix}$ and $A^2 = 3I$, where I is the identity matrix of order 2. Find x and y . ICSE 2020, 3 marks

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